

Dose-Finding Plates

Two identical plates from the same cells on the same day, one at 15 °C and one at 27 °C. No drug. They answer two questions at once: the working dose of each inhibitor, and whether the inhibitors cost growth only in the heat (the AOX-engagement signature). Repeat on two more days from fresh cultures = the figure.

plate DF_15_1 · 15 °C

plate DF_27_1 · 27 °C

no prothioconazole in cell wells

15 wells per plate

OD₆₀₀ 0.0005

GATE before anything goes in a vial

- 1. **Culture OD ≥ 0.10** and healthy under the scope. Below 0.10: wait a day. Never dilute-and-hope.
- 2. **Reader calibrated** (two-point) on the washed vials. One vial of plain air-shaken YMS must read ~7–8 mg/L. If not, fix that first.
- 3. nPG master made **fresh today**; SHAM master from the foil-wrapped stock. Both clear.

MAP same layout on both plates

	DF_15_1 15 °C incubator			DF_27_1 27 °C incubator		
	1	2	3	4	5	6
A	A1 CTRL EtOH	A2 CTRL EtOH	A3 nPG 0.0125 mM	A4 nPG 0.025 mM	A5 nPG 0.05 mM	A6 nPG 0.1 mM
B	B1 SHAM 0.0125 mM	B2 SHAM 0.025 mM	B3 SHAM 0.05 mM	B4 SHAM 0.1 mM	B5 SHAM 0.2 mM	B6
C	C1 BLANK nPG 0.1	C2 BLANK SHAM 0.2	C3 BLANK drug 4 mg/L	C4 BLANK drug 4 + nPG 0.1	C5	C6
D	D1	D2	D3	D4	D5	D6

 control (cells + ethanol)  nPG ladder  SHAM ladder  cell-free blanks (C1–C4)

C3–C4 are new: they show the drug itself doesn't move the sensor chemistry, so blank-subtraction in the main run is defensible. 11 cell wells + 4 blanks = 15 per plate, 30 total.

FILL every well the same, one thing changes

cell wells (A1–B5): 5.03 mL cells (OD 0.0005) + 20 μ L P0 (DMSO) + 10 μ L inhibitor stock (or plain EtOH for A1–A2)

blanks C1–C2: 5.03 mL sterile YMS + 20 μ L P0 + 10 μ L inhibitor

blanks C3–C4: 5.03 mL sterile YMS + 20 μ L P4 (1 mg/mL drug) + 10 μ L EtOH (C3) / 10 μ L 50 mM nPG (C4)

Cells: make **~130 mL at OD 0.0005** (mL culture = $0.0005 \times 130 \div$ culture OD, top up with YMS). Fill the two plates alternately from the same batch, swirling between draws, so both get the same cells. Foil everything with nPG/SHAM. Cap, invert gently, no bubbles.

STOCKS ethanol dilutions of your masters — 10 μ L/well, ~1 mL each covers both plates

FINAL	NPG STOCK (FROM 50 MM)	SHAM STOCK (FROM 250 MM)
0.0125 mM	6.25 mM · 125 μ L+875 EtOH	6.25 mM · 25 μ L+975 EtOH
0.025 mM	12.5 mM · 250 μ L+750 EtOH	12.5 mM · 50 μ L+950 EtOH
0.05 mM	25 mM · 500 μ L+500 EtOH	25 mM · 100 μ L+900 EtOH
0.1 mM	50 mM · master neat	50 mM · 200 μ L+800 EtOH
0.2 mM	– (nPG stops at 0.1)	100 mM · 400 μ L+600 EtOH

Not serial: every stock is made straight from its master. All doses sit 2.5–40 $\times$ below the 0.5 mM SHAM that abolished growth last run.

RUN

1. Both plates into their readers within the same hour. Dark.
2. Export names: **DF_15_1_Oxygen, DF_27_1_Oxygen**. Log culture OD, start time, calibration reading.
3. 27 $^{\circ}$ C turns over in ~4 days; 15 $^{\circ}$ C in ~5–6. Harvest by the curves, not the clock.
4. Same day, start the **induced flask** (27 $^{\circ}$ C + 2 mg/L drug, ~50 mL, inoculate ~OD 0.02) for the azide plate.

READ two answers from one pair of plates

The dose (27 °C plate)

Growth vs inhibitor dose. Working dose = highest well within ~15–20% of A1–A2, confirmed later by the azide plate as actually blocking. Carry **two** doses into the main run.

The split (both plates)

Same ladder, two temperatures. Inhibitor costing growth at 27 °C but not at 15 °C = AOX engaged only in the heat. Replicated on three days, this is a figure on its own.

Replicates 2 and 3: fresh culture, same two plates, next two weeks. Rotate which row holds which ladder between replicates.